VERIFICATION REPORT TITLE

DAK SRONG HYDROPOWER PROJECT

Document Prepared By

(Carbon Check (India) Private Ltd)

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Summary:

A brief description of the verification and the project

***Verification**: Kyoto Energy Pte Ltd. has appointed Carbon Check (India) Private Ltd., to carry out the second periodic verification of the registered VCS project "Dak Srong 2 Hydropower Project" (VCS ID: 971), with regards to the relevant requirements of VCS Standard Version 4.1 (dated 22/04/2021).*

***Project:** The project "Dak Srong 2 Hydropower Project", is a large-scale project activity which employs approved CDM methodology ACM0002 version 11, "Grid connected energy generation from renewable sources."/B02/ The purpose of the project activity is to generate electricity from the renewable energy which is distributed to the national grid. The installed capacity of the project is 24MW and the total net electricity generation during this monitoring period is about 384,854MWh. Total emission reduction amount due to the project activity during the second monitoring period (01/03/2012-22/02/2018) is 186,159 tCO_2/02/. The project is located at Yang Nam, Ya Ma, and Dak Hninh Communes,Kong Chro district and the GPS coordinates of Dak Srong 2 Hydropower Project is: Latitude of 13^0 41'45"N and longitude of 108^0 33'30"E.*

***Purpose**: The purpose of the verification is to review the monitoring results and verify that monitoring methodology was implemented according to monitoring plan and monitoring data, used to confirm the reductions in anthropogenic emissions by sources is sufficient, definitive, and presented in a concise and transparent manner. In particular, monitoring plan, monitoring report and the project's compliance with relevant VCS, UNFCCC and host Party criteria are verified in order to confirm that the project has been implemented in accordance with previously registered design and conservative assumptions, as documented.*

***Scope:** The verification scope is defined as an independent and objective review of the monitoring report (MR) against the relevant criteria and guidance documents provided by VCS which include the following: VCS Program Guide (v4.0), VCS Standard (v4.1, dated 22/04/2021), Program Definitions (v4.1), Registration & Issuance Process (v4.0, dated 19/09/2019) VCS Validation and Verification Manual (v3.2, dated 19/10/2016) applicable at the time in order to confirm the emission reductions produced during the monitoring period are in accordance with the project activity as provided in the registered VCS PD and CDM PDD/B03/. The CDM approved methodology ACM0002, Version 11/B02/ has been applied for the project activity.*

The method and criteria used for verification

The verification consists of the following four phases:

I. A desk review of the project description documents

- *A review of data and information.*

Cross checks between information provided in monitoring report and information from sources with all necessary means without limitations to the information provided by the project proponent.

II. Remote Audit

- *Interviews with relevant stakeholders in host country with personnel having knowledge with the project development via telephone, email.*
- *Cross checking between information provided by interviewed personnel with all necessary means without limitations to the information provided by the project proponent.*

III. Reference to available information relating to projects or technologies similar to project under verification and review based on the approved methodology being applied for the appropriateness of formulae and accuracy of calculations.

IV. The resolution of outstanding issues and the issuance of the final verification report and opinion.

The number of findings raised during verification

During the course of verification, a total of four (08) findings were raised, which include:

- *Corrective Action Requests (CARs): (05)*
- *Clarification Requests (CLs): (03)*
- *Forward Action requests (FARs): Nil (00)*

All the raised findings have been successfully closed.

Any uncertainties associated with the verification

There are no uncertainties associated with the verification of the project activity. The verification has been done with a reasonable level of assurance.

Summary of the verification conclusion

Carbon Check (India) Private Ltd. concludes the verification with a positive opinion that the VCS large scale project activity "Dak Srong 2 Hydropower Project" as described in the monitoring report (version 2.3 dated 05/04/2022)/01-c/,registered CDM PDD registered VCS PD (version 04, dated 09/02/2012) /B03/, meets all applicable VCS requirements, including those specified in the VCS Standard (v4.1, dated 22/04/2021), relevant methodology, tools and guidelines.

The project activity and its emission reduction calculation were correctly implemented and meets all requirement for the VCS standard and guidelines and correctly applies the baseline and monitoring methodology ACM0002 version 11, "Grid connected energy generation from renewable sources. The monitoring system is in place and the emission reductions are calculated without material misstatement Carbon Check (India) Private Ltd. therefore requests the issuance of the project as a VCS project activity, with a reasonable level of assurance

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1 INTRODUCTION

1.1 Objective

Kyoto Energy Pte Ltd. (Project Proponent) has appointed the VVB, CCIPL for verification service for the VCS registered project activity - "Dak Srong 2 Hydropower Project" located at the Gia Lai Province, Yang Nam, Ya Ma, and Dak Hninh Communes, Kong Chro district. Vietnam.

Verification is the periodic independent review and ex post determination of both quantitative and qualitative information by a Validation and Verification Body (VVB) of the monitored reductions in GHG emissions that have occurred as a result of the VCS project activity during a defined monitoring period.

The purpose of verification is to review the monitoring results and verify that the monitoring methodology was implemented according to the monitoring plan and monitoring data and used to confirm the reductions in emissions is sufficient, definitive and presented in a concise and transparent manner. Carbon Check's objective is to perform a thorough, independent assessment of the registered projects activities. In particular the, monitoring plan, monitoring report and the project's compliance are verified against the relevant criteria and guidance documents provided by VCS. This allows for the confirmation that the project has been implemented in accordance with the VCS registered VCS PD/B03/ and conservative assumptions, as documented. And, also to confirm if the monitoring plan is in compliance with the VCS PD/B03/ and approved monitoring methodology/B02/, The baseline is established using the approved methodology ACM0002 (version 11), "Grid connected energy generation from renewable sources." /B02/This methodology refers to the latest. The objective of this verification was to verify and certify emission reductions reported for the "Dak Srong 2 Hydropower Project" for the period 01/03/2012 to 22/02/2018

1.2 Scope and Criteria

The verification of this project is based on the registered VCS Project Description, CDM PDD/B03/, the Monitoring Report of this monitoring period /01-c/, emission reduction calculation spread sheet /02/, supporting documents made available to the verifier and information collected through performing interviews and during the on-site assessment. Furthermore, publicly available information was considered as far as available and required.

Carbon Check has employed a risk-based approach in the verification, focusing on the identification of significant risks and reliability of project monitoring and generation of emission reductions.

The verification is carried out on the basis of the following requirements (latest available on VCS website at the time of verification), applicable for this project activity:

- *VCS Program Guide (v4.0, dated 19/09/2019)*
- *VCS Standard (v4.1, dated 22/04/2021)/B01 a/*
- *Program Definitions (v4.1, dated 22/04/2021) /B01 e/*
- *Registration & Issuance Process (v4.0, dated 19/09/2019) /B01 d/*
- *VCS Validation and Verification Manual (v 3.2, dated 19/10/2016)/B01 c/*
- *CDM Methodology: ACM0002: Grid-connected electricity generation from renewable sources (Version 12.1.0) /B02/*
- *Other relevant rules, including the host country legislation*

The scope of this verification, by independent checking of objective evidence, is as follows:

- *To verify that the project is implemented as described in the project description*
- *To assess the project's compliance with other relevant rules including the host country legislation.*
- *To assess the implementation of the monitoring plan content as mentioned in the the VCS-PD/B03/*
- *To confirm that the monitoring system is implemented and fully functional to generate voluntary emission reductions (VERs/VCUs) without any double counting and*
- *To establish that the data reported are accurate, complete, consistent, transparent, and free of material error or omission by checking the monitoring records and the emissions reduction calculation /02/.*
- *To evaluate the GHG emission reduction data and express a conclusion with a reasonable level of assurance about whether the reported GHG emission reduction data is free from material misstatement.*
- *To verify that reported GHG emission data is sufficiently supported by evidence.*
- *The verification shall ensure that the reported emission reductions are complete and accurate to be certified.*

The method and criteria used for verification consisted of the following phases:

1) *Completeness check and desk review:*
2) *Remote Audit.*
3) *Resolution of outstanding issues and issuance of final verification report and applicable VCS Validation and Verification Deeds of Representation.*

Carbon Check (India) Private Ltd. conducts all its work under strict rules to safeguard impartiality and ensure the independence of the verification team. The verification does not provide any consulting or recommendations for the client. However, stated requests for clarifications and/or corrective actions may provide input for improvement of the monitoring activities.

1.3 Level of Assurance

Verification team confirms that all relevant assumptions and data are listed in the MR, including their references and sources and that all data and parameter values used in the MR are considered reasonable in the context of the project and all estimates of the baseline emissions can be replicated using the data and parameter values provided in the MR and PD of the project. No uncertainties associated with the calculations of emissions have been observed by the verification team. Thus, the project achieves a reasonable level of assurance.

☒ *Reasonable level of assurance*

☐ *Limited level of assurance*

1.4 Summary Description of the Project

The Project is a run of river hydro power project using the water from small run of river reservoir, and consists of a weir, a penstock, a powerhouse (containing turbines and generators) and a tailrace. Project has an installed capacity of 24 MW produced by three units (each 8 MW).. The electricity generated by the project activity reduces emissions. With a total net electricity generation of about 384,854 MWh during the Monitoring Period, and an emission factor of 0.5764 tCO_2/MWh, total emission reductions amount to 186,159 tCO_2 for this monitoring period. The project commercial operation date is 30/10/2010. The expected operational lifetime of the project activity is 30 years. The crediting period lasts 10 year and is renewable twice. The first crediting period is started on 30/10/2010. The second monitoring period is in between 01/03/2012 to 22/02/2018.

The total estimated GHG emission reductions expected from the project activity are 266,065 tCO_2e for this monitoring period and average of 44,466tCO2/year.

The total emission reductions for the reported monitoring period 01/03/2012 to 22/02/2018 are 186,159 tCO_2e

Verification team confirms that there will be no double counting occur and the same has been verified on the basis of review of the CDM project page, GCC, GS and other standards and also a declaration of no double counting has also been submitted by the PP./21/

2 VERIFICATION PROCESS

2.1 Method and Criteria

The method and criteria used for verification:

The verification consists of the following three phases:

1) *Completeness check and desk review of the validation report, monitoring plan, monitoring report, monitoring methodology, VCS PD, /B03/ applicable tools in particular attention to the frequency of measurements, quality of metering equipment's including calibration requirements, QA/QC procedures and other relevant documents.*

2) *Remote audit interviews (including follow-up interviews with project stakeholders, when deemed necessary). The remote audit assignment includes the following:*
   - *An assignment of implementation and operation of project activity with respect to validated VCS PD;/B03/ and MR /02/*
   - *Review of information flows for generating, aggregating, and reporting the monitoring parameters.*
   - *Interview with relevant personals to determine whether the operational and data collection procedures are implemented and in accordance with monitoring plan of the validated VCS PD /B03/.*
   - *Cross check of information and data provided in the monitoring report with plant logbooks, inventories, purchase records or similar data sources.*
   - *Check of monitoring equipment's, calibration frequency and monitoring practice in-line with methodology and validated VCS PD/B03/.*
   - *Review of assumptions made in calculating the emission reduction.*
   - *Implementation of QA/QC procedure in-line with the validated VCS PD and methodology requirement.*
3) *Resolution of outstanding issues and the issuance of the final Verification report and if applicable, the VCS Validation and Verification Deeds of Representation*

2.2 Document Review

The registered VCS PD,CDM PDD /B03/, VCS MR /01-c/ emission reduction calculation spread sheet/02/ and supporting documents related to the project implementation, project design, monitoring and baseline were reviewed as per VCS standard (version 4.1) standard requirements. The desk review included:

A review of the data and information presented to verify completeness and consistency in accordance with VCS standard (version 4.1) requirements/B01/.

A review of the approved monitoring plan and monitoring methodology, paying particular attention to the frequency of measurements, quality of monitoring equipment (including calibration requirements) and the quality assurance and quality control (QA/QC) procedures.

An evaluation of data management and the QA/QC system in the context of their influence on the generation and reporting of emission reductions.

Data input values were also checked from the records maintained by the project proponents. Results of calculations reported in the monitoring report were checked against data values as available from the project proponent in VER calculation sheet /02/.

These data values and other information related to project performance are available in the form of data records duly archived and maintained as per the quality assurance/quality control procedure specified as a part of monitoring plan in the registered /VCS-PD/B03/.

Furthermore, the verification team used additional documentation by third parties like host-party legislation, technical reports referring to the project design or to the basic conditions and technical data.

The Monitoring report version 01 dated 19/11/2021 /01-a/ was initially reviewed and CCIPL requested the PP to present the supporting information and documents. The documents reviewed by CCIPL are listed below in Appendix 1. Through the process of the verification, the revised monitoring report and the supporting documents were evaluated to confirm the actions taken by the PP to the CARs and CLs issued by the verification team.

2.3 Interviews

The table below describes the remote interview process and further identifies personnel, including their roles, who were interviewed and/or provided information additional to that provided in the project description /B03/ and any supporting documents.

	Date	Name	Organisation	Topic
/a/	*09/12/2021*	*Ms Lan Nguyen Thi*	*Tay Nguyen Hydropower Company Limited*	• *Project Design* • *Project Implementation status* • *Project start date and Project Location* • *Roles and responsibility, Project Implementationand Operation status, Qualification and Training, Roles and responsibility.* • *Involved in Emissions Trading Programs and Other Binding Limits or Other Forms of Environmental Credit.* • *Impact of the capacity change on the applicability of the methodology, additionality, or the appropriateness of the baseline scenario.*

/b/	09/12/2021	Nhan Le Trang	Tay Nguyen Hydropower Company Limited	• Project design • Turbine details • Monitoring plan • Monitoring proceduresand monitoring equipment • Maintenance and operation plan • Data and information flow, Data input device, Roles and responsibility, Project implementation and operation, monitoring procedure.
/c/	09/12/2021	Abraham Antony	Kyoto Energy Pte. Ltd.	• VCU calculation and completeness of monitoring report, • Project implementationand operation, Project design, monitoringprocedure, data and information flow, compliance of monitoring plan with monitoring methodology and VCS-PD. • Project Implementation status

2.4 Site Inspections

Carbon Check has not conducted an on-site inspection due to the recent pandemic COVID19 and due to its related policy measures created restrictions all over the world impacting travel activities on an international level and even for in-country travel. A reasonable level of assurance has been maintained through the alternative means used for the purpose of validation as follows:

- *An assessment of the implementation and operation of the proposed grouped project through remote interviews with the representatives of project proponent.*
- *Assessment of the project design, turbines details, monitoring plan, monitoring procedures and monitoring equipment.*
- *Assessment of the design change impacts on the applicability of the methodology,additionality, or the appropriateness of the baseline scenario*

- Confirmation of the applicability of the methodology and monitoring and controllinginstruments and operational arrangements.

2.5 Resolution of Findings

This section summarizes the findings from the verification of the project activity. In this section the findings from the document review, assessments and remote interviews are provided.

Material discrepancies identified in the course of the verification are addressed either as CARs, CLs or FARs.

Corrective action requests (CAR) are issued, where:

i. Mistakes have been made with a direct influence on project results requiring adjustments of the VERs/VCUs monitoring report.

ii. Applicable methodological specific requirements have not been met.

A Clarification request (CL) may be used where additional information is needed to fully clarify an issue or where the information is not transparent enough to establish whether a requirement is met.

A total 05 CAR, and 03 CLs had been raised by the verification team. Please refer to Appendix 4 below for the details of the CARs/CLs and their closure.

2.5.1 Forward Action Requests

A forward action request (FAR) should be issued, where:

i. the actual project monitoring and reporting practices requires attention and /or adjustment for the next consecutive verification period, or

ii. an adjustment of the MP is recommended.

In the context of FARs, risks have been identified, which may endanger the delivery of high-quality emissions reductions in the future, i.e. by deviations from standard procedures as defined by the MP. As a consequence, such aspects should receive a special focus during the next consecutive verification. A FAR may originate from lack of data sustaining claimed emission reductions.

No FAR was raised during first verification neither FAR has been raised during this monitoring period.

2.6 Eligibility for Validation Activities

Validation/Verification body (VVB), Carbon Check (India) Private Ltd. holds accreditation for verification for the relevant sectoral scope 1 and is eligible for validation/verification for the project activity.

Please refer: https://verra.org/project/carbon-check-india-private-ltd/

3 VALIDATION FINDINGS

3.1 Participation under Other GHG Programs

The project is registered under the clean development mechanism with the Project title: Dak Srong 2 Hydropower Project, and UNFCCC reference number 3389. The project was registered on 23/02/2011. The date of issuance of the project is 15/11/2012 and for the monitoring period 23/11/2011 to 29/02/2012 the amount of Certified Emission Reduction was 41,560 tCO_2. This detail was cross verified by the verification team on the basis of review of registered CDM PDD/B03/, MR of 1st monitoring period /B04/ and corresponding Verification report/B04/ and the CDM project page.

3.2 Methodology Deviations

Not applicable as project does not involve any methodology deviation.

3.3 Project Description Deviations

Not applicable as project does not involve any deviation.

3.4 Grouped Project

Not applicable as the project is not a grouped project.

4 VERIFICATION FINDINGS

4.1 Project Implementation Status

The project activity "Dak Srong Hydropower Project" a run of river hydro power plant was registered as VCS project having VCS ID 971 applying the methodology ACM0002 version 11.0. "Grid-connected energy generation from renewable sources /B02/. The Project Proponent for *the project activity is Tay Nguyen Hydropower Company Limited and the same is verified from the power purchase agreement /15/*

The project activity involves the generation of the electricity through hydro power plant with total generation capacity of 24MW/01-c/ made up of three units each with the installed capacity of 8 MW. (Verified from the rated capacity of generators provided in the technical specification of the equipment. /03-c/

The hydropower plant uses three turbines, each having a capacity of 8290.2KW, to convert the kinetic energy into electricity. The type of turbine used is Francis HLA 5510-LO-176 with net head 39.28m and net discharge rate 24.11 m^2/sec */03-a/.*

There are three generators in the plant, each generator has 8000KW producing power capacity with rated voltage 6300V. The electricity generated by the project will be delivered to the VietNanam national grid via a new single -circuit 110KV transmission line. The energy meters of each unit measure the export and import of the electricity generated. The representatives from EVN (Viet Nam national grid; Central Power Corporation), PP and Gia Lai province electricity company take down the monthly reading on first day of every month. This is then recorded in the EVN receipt and is signed by these representatives. Invoices are raised by PP based on these receipts and accordingly payment is done by EVN. EVN further cross checks the main meter reading with back up meter reading so as to ensure that the reading recorded from the main meter is correct. Verification team during the site visit checked the same and confirms that the same procedure is followed at site was confirmed during remote audit.

Connection point details: The electricity generated from Dak Srong 2 hydropower plant will be connected to the National Grid through a new 110 kV single circuit transmission line from 110 kV Transformer Station of the Dak Srong 2 to 110 kV transformer station of the Dak Srong 2 A hydropower plant and then connect through the 110 kV transformer station of Dak Srong hydropower plant to the National Grid

The project boundary includes the electricity generation equipment at the site "Dak Srong Hydropower Project" and the national grid of the host country, VietNam.

The ex-ante fixed grid emission factor of 0.5764 tCO_2/MWh *has been used for the baseline emissions calculations, which is in line with the registered VCS PD. /B03/*

The registered PD clearly describes the monitoring plan and responsibility of monitoring. These data are subject to the accounting quality systems of both parties, i.e., from the power purchase of the project owner and the Central Power Company. /15/ With this, no additional structures or processes have to be implemented to insure the availability of necessary data for monitoring. During the site visit, monitoring and reporting procedures were confirmed with the relevant staff and through the document review.

The monitoring plan is in accordance with the UNFCCC approved methodology ACM0002, version 11. All the data is collected and archived in accordance with the methodology and included in *the monitoring plan. The monitoring has been carried out in accordance with the provision of monitoring plan, the verification team reviewed if*

- *The monitoring of reductions in GHG emissions resulting from the proposed VCS project activity were implemented in accordance with the monitoring plan contained in the registered VCS-PD /B03/*
- *The monitoring plan and the applied methodologies had been properly implemented and followed by the project participants.*
- *All parameters stated in the monitoring plan, the applied methodologies and relevant VCS decisions had been sufficiently monitored and updated.*
- *The responsibilities and authorities for monitoring and reporting were in accordance with the responsibilities and authorities stated in the monitoring plan.*

The GHG emission reductions generated by the project activity are not included in an emissions trading program or any other mechanism that includes GHG allowance trading, this has been confirmed from the review of the PD and also confirmed during the site visit. The project activity has not received or sought any other form of environmental credit, this has been confirmed from the review of the PD and also confirmed during the site visit.

The sustainable development contribution of the project activity to the host country, VietNam have been provided in the MR and the contributions were confirmed with the PP during the remote audit. The project has positive impacts with respect to the environment (offsetting fossil fuel use and lowering greenhouse gas emissions), society (providing jobs, development of roads), technologically (technology transfer) and economy (satisfying growing energy demands to allow the country and region to develop and alleviate poverty). Also, PP has voluntarily agreed to donate a portion of CER revenue from the project for sustainable development initiatives.

Overall, the project has been implemented in accordance with the registered VCS PD /B03/

4.2 Safeguards

4.2.1 No Net Harm

The project Dak Srong 2 have done its environmental impact assessment as per the Vietnamese law for the Dak Srong 2 project, the reservoir volume is 85.6x106 m3 (exceeding 106 m3 as required in Decree 80/2006/ND-CP), This EIA has been approved by the relevant local authorities, Gia Lai Provincial People's Committee, dated 31-January-2008. Verification team on the basis of review of the monitoring records namely environmental monitoring reports for all the four quarters, environmental protection report 2020_Dak Srong and Test results Dak Srong 2 (Quarter I, II, III, IV) /19/ for this monitoring period maintained by PP in compliance with the EIA report and registered PD confirms that the PP have followed the environmental monitoring plan and complies with the related Vietnamese law.

4.2.2 Local Stakeholder Consultation

The stakeholder consultation meeting was held at the Administrative Building of Kong Ch'ro District People's Committee, Gia Lai Province, Viet Nam at 9:00 AM on 28-July-2008. Verification team has reviewed the supporting document namely Water usage consultation 2&2A /19/which is a report for a recent consultation with community leaders and specifically addressing water usage. The report shows that there is no reported issues nor any pending/historical issues and thus confirms that there is no grievance registered for this monitoring period.

4.3 AFOLU-Specific Safeguards

Not applicable as this is non AFOUL

4.4 Accuracy of GHG Emission Reduction and Removal Calculations

*The **baseline emissions** are to be calculated as follows:*

$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$	*Eqn. (1)*

Where:

BE_y	=	*Baseline emissions in year y (tCO_2/yr)*
$EG_{PJ,y}$	=	*Quantity of Net electricity generation that is produced and fed into the grid as a result of the implementation of the VCS project activity in year y (MWh/yr)*
$EF_{grid,CM,y}$	=	*Combined margin CO_2 emission factor for grid connected power generation in year y calculated using the latest version of the "Tool to calculate the emission factor for an electricity system" (tCO_2/MWh)*
y	=	*A given period (year)*

The project activity is the installation of a new grid-connected renewable power plant, thus:

$EG_{PJ,y} = EG_{facility,y}$	*Eqn. (2)*

Where:

$EG_{PJ,y}$	=	*Quantity of Net electricity generation that is produced and fed into the grid as a result of the implementation of the VCS project activity in year y (MWh/yr)*
$EG_{facility,y}$	=	*Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)*

Then equation (1) becomes:

$BE_y = EG_{facility,y} \times EF_{grid,CM,y}$	*Eqn. (3)*

Where:

BE_y	=	*Baseline emissions in year y (tCO_2/yr)*
$EG_{facility,y}$	=	*Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)*
$EF_{grid,CM,y}$	=	*Combined margin CO_2 emission factor for grid connected power generation in year y calculated using the latest version of the "Tool to calculate the emission factor for an electricity system" (tCO_2/MWh)*
y	=	*A given period (year)*

$EF_{grid,CM,y}$ was calculated at the time of validation as 0.5764 tCO_2/MWh.

***Project emission** are to be calculated as:*

As per the ACM0002 (v11.0), project emission from reservoir is calculated as below,.

$$PD = \frac{Cap_{PJ} - Cap_{BL}}{A_{PJ} - A_{BL}}$$ Equation (1)

Where:

PE_y	=	*Project emissions in year y (tCO_2/yr)*
$PE_{HP,y}$	=	*Project emissions from water reservoirs of hydro power plants in year y (tCO_2e/yr)*

As calculated in section 5.2 of the MR the project density is 5.27 which is greater than 4 W/m2 and less than or equal to 10 W/m2 the project emission can be calculated as :-

$$PE_{HP,y} = \frac{EF_{Res} \times TEG_y}{1000}$$ Equation (2)

As per the calculation in section 5.2 of MR the projects emissions are 35,671 tCO_2.

Leakage:

No leakage is applicable for the project as per the applied methodology.

$LE_y = 0$	*Eqn. (5)*

Where:

LE_y	=	*Leakage emissions in year y (tCO_2/yr)*

Net GHG Emission Reduction and Removals

Emission reductions related to project activities must be estimated using ACM0002 as:

$ER_y = BE_y - PE_y - LE_y$	*Eqn. (11) of the methodology*

Where:

ER_y	=	*Emission Reductiom in year y (tCO_2/yr)*
BE_y	=	*Baseline emissions in year y (tCO_2/yr)*
PE_y	=	*Project emissions in year y (tCO_2/yr)*
LE_y	=	*Leakage emissions in year y (tCO_2/yr)*

On applying the values of baseline emissions, project emissions and leakage emissions as mentioned in table under section 5.4 of the MR the calculated value of emission reduction is 186,159 tCO_2 for this monitoring period.

The verification team has determined whether the registered monitoring plan in the PD /B03/ has been properly implemented and followed by the PP and whether all parameters fixed ex-ante for emission reduction calculation are as per the registered PDD /B04/. The verification team's assessment of each data and parameter fixed ex-ante is provided in Appendix 5 of this report.

The values are consistent with the registered VCS PD/B03/ and fixed ex-ante during 1st renewable crediting period of the project activity. The fixed ex-ante data and parameter have been listed in the monitoring report/01-c/ and confirmed by the verification team as correct and consistent with that stated in the registered VCS PD /B03/. The verification team confirms that the MR/01-c/ and the ER calculation spreadsheet/02/ have considered the parameters fixed ex-ante correctly.

The details of the data and parameters monitored during this monitoring period is described in the section 4.2 of the Monitoring report /01 -b/.

Verification team confirms that that the value of the data provided for the monitored parameter is in compliance with the registered monitoring plan as provided in the registered VCS PD/B04/. However verification team has observed the incremental change in the actual generation as compared to the estimated generation in the year 2017 and few other months has been mentioned in the CL 03 in the appendix 4.

4.5 Quality of Evidence to Determine GHG Emission Reductions and Removals

CCIPL was able to confirm that the calculations are based on authentic data. The spreadsheets /02/used to calculate the VCUs calculations and all figures were tracked, checked and found to be consistent.

The quality of supporting evidence submitted to the VVB for verification is adequate and found to be verifiable. The transfer of carbon rights and other supporting documents related to quality and maintenance were checked by the verification team during the remote audit to confirm the authenticity of the documents and to check the correctness of the calculation.

When verifying the reported emission reductions, CCIPL ensured that there was a clear audit trail that contained the evidence and records that validate the stated figures. All source documents that form the basis for assumptions and other information underlying the GHG data were checked by the verification team.

When assessing the audit trails, CCIPL also examined:

1. *Whether sufficient evidence was available, both in terms of frequency and in covering the full monitoring period*
2. *The source and nature of the evidence.*
3. *If comparable information was available from sources other than that used in the monitoring report, CCIPL cross-checked the monitoring report against the other sources to confirm that the stated figures were correct.*

CCIPL also assessed that the data collection system met the requirements of the monitoring plan as per the applied methodology /B02/.

Proper data management inclusive of data acquisition and aggregation, data management system is being followed for the project activity.

The monitoring personnel at site are well trained and follow reproducible routines. Thus, they are competent to carry out the relevant tasks with sufficient accuracy

Assessment of Ex-post Monitored Parameters:-

Monitoring Parameter Requirement	Assessment/ Observation by the DOE
Data / Parameter: (as in monitoring plan of PDD):	$EG_{facility,y}$
Description of parameter	*Quantity of net electricity generation supplied by the project plant/unit to the grid in year y*
Data unit	*MWh*

Reported value	*384,854MWh*
Measuring frequency	*Continuous measuring*
Recording frequency	*Monthly reading*
Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	*Yes*
Type of monitoring equipment	*Bidirectional Electricity Meters installed at theDak Srong 2 Hydroelectric power plant). The configuration for all the meters is as provided in the appendix 1 of the VCS MR /01-c/* *The verification team has cross checked the information provided in the appendix 1 of the MR /01-c/from the calibrations certificate submitted by the PP. and remote audit interview with the plant person, thus verification team confirms that the information provided regarding the meter type, accuracy class, serial number and calibration records by the PP are appropriate and deemed acceptable to the verification team.*
Is accuracy of the monitoring equipment as stated in the PDD?	*Yes, the accuracy of the monitoring equipment is as stated in the VCS PD/B03 /.*
Calibration frequency /interval	*As per the Monitoring report Periodic calibration of the meters are done every 2 years and same was cross checked by the review of the copy of decision 25/2007/QD-BKHCN/20/ [provided by the PP}.*
Company performing the calibration	*Central power company*
Did calibration confirm proper functioning of monitoring equipment? (Yes / No)	*Yes*
Is(are) calibration(s) valid for the whole reporting period?	*Yes*
If applicable, has the reported data been crosschecked with other available data?	*Yes, the reported data has been cross-checked with the electricity sales invoices/08/ and /07/*
How were the values in the monitoring report verified	*The values in the monitoring report were verified through the comparison with the values in the ER sheet/02/ and the raw data provided therein, and cross checked with the electricity sales invoices/08/*
Does the data management (from monitoring equipment to emission reduction calculation) ensure correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	*Yes, the data management ensures correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place.*

Monitoring Parameter Requirement	Assessment/ Observation by the DOE
Data / Parameter: *(as in monitoring plan of PDD):*	*TEG_y*
Description of parameter	*Total electricity produced by the project activity, including the electricity supplied to the grid and the electricity supplied to internal loads*
Data unit	*MWh*
Reported value	*396,342MWh*
Measuring frequency	*Continuous measuring*
Recording frequency	*Monthly reading*
Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	*Yes*
Type of monitoring equipment	*TEG1, TEG2, and TEG3* *The configuration for all the meters is as provided in the appendix 1 of the VCS MR /01-c/* *The verification team has cross checked the information provided in the appendix 1 of the MR /01-c/from the calibrations certificate submitted by the PP. and remote audit interview with the plant person, thus verification team confirms that the information provided regarding the meter type, accuracy class, serial number and calibration records by the PP are appropriate and deemed acceptable to the verification team.* *Moreover, it was observed by the verification team that there was delay in the calibration check of the TEG meters by 5 days in year 2016 as stated in Appendix 1 of the MR*. *And thus, due to delay in calibration a penalty (increase) of 0.5% has been applied to the total generation for the month of October 2016.* *The results of delayed calibrations reveal that the meters did not have errors in excess of the maximum permissible error limit (accuracy class) of 0.5%. Thus, the application of a penalty of 0.5% is deemed as a conservative course of action. In addition, the application of the error over the whole month, rather than for the 5 delayed days of calibration, is also accepted as a conservative option.*
Is accuracy of the monitoring equipment as stated in the PDD?	*Yes, the accuracy of the monitoring equipment is as stated in the VCS PD/B03 /.*
Calibration frequency /interval	*As per the Monitoring report Periodic calibration of the meters are done every 2 years and same was cross checked by the review of the copy of decision 25/2007/QD-BKHCN/20/ [provided by the PP}.*

Company performing the calibration	*Central power company*
Did calibration confirm proper functioning of monitoring equipment? (Yes / No)	*Yes*
Is(are) calibration(s) valid for the whole reporting period?	*Yes*
If applicable, has the reported data been crosschecked with other available data?	*Yes, the reported data has been cross-checked with the electricity sales invoices/08/ and /07/*
How were the values in the monitoring report verified	*The values in the monitoring report were verified through the comparison with the values in the ER sheet/02/ and the raw data provided therein, and cross checked with the electricity sales invoices/08/*
Does the data management (from monitoring equipment to emission reduction calculation) ensure correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	*Yes, the data management ensures correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place.*

Monitoring Parameter Requirement	Assessment/ Observation by the DOE
Data / Parameter: (as in monitoring plan of PDD):	A_{PJ}
Description of parameter	*Area of the reservoir measured with topographic drawings in the surface of the water, after implementation of the project activity, when reservoir full*
Data unit	m2
Reported value	4,550,000 m^2 Water level measured on site a water level gauge, and surface area is calculated by area to water level chart (made specifically by a third-party consultant for the reservoir associated with the project activity).
Measuring frequency	*Yearly*
Recording frequency	*Yearly*
Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	*Yes*
Type of monitoring equipment	*N/A*

Is accuracy of the monitoring equipment as stated in the PDD?	*N/A*
Calibration frequency /interval	*N/A*
Company performing the calibration	*N/A*
Did calibration confirm proper functioning of monitoring equipment? (Yes / No)	*N/A*
Is(are) calibration(s) valid for the whole reporting period?	*N/A*
If applicable, has the reported data been crosschecked with other available data?	*N/A*
How were the values in the monitoring report verified	*By cross check surface area calculation by third party consultant for the reservoir of the project and based on topographical maps./17/*
Does the data management (from monitoring equipment to emission reduction calculation) ensure correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	*N/A*

Monitoring Parameter Requirement	Assessment/ Observation by the DOE
Data / Parameter: *(as in monitoring plan of PDD):*	*CAP BJ*
Description of parameter	*Installed capacity of the hydro power plant after the implementation of the project activity*
Data unit	W
Reported value	24000 m^2
Measuring frequency	*yearly*
Recording frequency	*continuously*
Is measuring and reporting frequency in accordance with the monitoring plan and monitoring methodology? (Yes / No)	*Yes*
Type of monitoring equipment	*N/A*
Is accuracy of the monitoring equipment as stated in the PDD?	*N/A*
Calibration frequency /interval	*N/A*
Company performing the calibration	*N/A*

Did calibration confirm proper functioning of monitoring equipment? (Yes / No)	*N/A*
Is(are) calibration(s) valid for the whole reporting period?	*N/A*
If applicable, has the reported data been crosschecked with other available data?	*N/A*
How were the values in the monitoring report verified	*By cross check from Name plates of generating equipment/18/*
Does the data management (from monitoring equipment to emission reduction calculation) ensure correct transfer of data and reporting of emission reductions and are necessary QA/QC processes in place?	*N/A*

4.6 Non-Permanence Risk Analysis

Not Applicable.

5 VERIFICATION CONCLUSION

Carbon Check (India) Private Ltd., (the VVB) has performed the second (2nd) emission reduction verification of the registered VCS project activity "Dak Srong 2 Hydropower Project" having (VCS ID.: 971). The Project Activity involves installation and operation of a new grid connected 24 MW hydropower plant consisting of three turbines located at Yang Nam, Ya Ma, and Dak Hninh Communes,Kong Chro district, Gia Lai Province Socialist Republic of Viet Nam . The project generates electricity from renewable source of energy (water) and generated electricity is supplied to the Vietnamese National grid. This will reduce and replace the equivalent amount electricity generated from the carbon intensive power plants in the VietNam national grid thus helps in reducing the GHG emissions.

The project owner is responsible for collection of data in accordance with the monitoring plan and the reporting of GHG emission reductions. It is VVB's responsibility to express an independent emission reduction verification opinion and certification statement on the reported GHG emission reductions from the project activity. The VVB does not express any opinion on the selected baseline scenario or on the validated and registered PD /B07/. The emission reduction verification is carried out in-line with the requirements of VCS Standard (v4.1) /B01-a/ and VCS Validation and Verification Manual (v3.2) /B01-c/.

The project verification was performed to identify the compliance with implementation and monitoring requirements, and to verify the actual amount of achieved emission reductions, through obtaining evidence and information during remote audit that included:

- *(i) checking whether the provisions of the monitoring methodology and the monitoring plan were consistently and appropriately applied and*
- *(ii) the collection of evidence supporting the reported data.*

The emission reduction verification is based on:

- *Registered CDM PDD (version 03.6; dated 19/06/2012) /B03/*
- *Registered VCS PD (version 1.1 dated 04/12/2012)/B03/*
- *Project Validation Report corresponding to CDM revised PD and VCS PD*
- *ACM0002 "Grid-connected electricity generation from renewable sources" (version 11.0) /B02/*
- *Monitoring Report (version 01)/01-a/, Monitoring Report (version 02)/01-b/*
- *Monitoring Report (version 02.1) dated 08/01/2022/01-c/*
- *Monitoring report (version 02.3 dated 05/04/2022/01-d/*

This statement covers emission reduction verification period from 01/03/2012 to 22/ 02/2018 (including both dates).

The VVB had raised 03 Clarifications (CLs), 05 Corrective Action Request (CARs) during the current emission reduction verification and zero (00) Forward Action Request (FAR)) have been raised during this emission reduction verification and have been satisfactorily resolved by the project proponent

The VVB considers it necessary to give reasonable assurance that the reported GHG emission reductions were calculated correctly on the basis of the approved baseline and monitoring methodology and the monitoring plan contained in the registered VCS PD /B03/ are fairly stated.

The conclusions can be summarised as follows:

- *The project is implemented and installed as planned and described in the registered VCS PD /B03/ and project activity confirms with the verification criteria for project and their GHG emission reductions or removals set out in the VCS rules.*
- *The monitoring plan is in accordance with the applied approved methodology, i.e., ACM0002, version 11.0/B02/ and monitoring plan as sought out in the registered VCS-PD/B03/.*
- *The monitoring system is in place and functional. The project has generated verifiable GHG emission reductions.*

As the result of the verification of the project activity, the verifier confirms that the GHG emission reductions are calculated without material misstatements in a conservative and appropriate manner. Carbon Check (India) Private Ltd. herewith confirms that the project has achieved emission reductions in the below mentioned reporting period as follows. The project complies with the verification criteria for projects and their GHG emissions reductions or removals set out in VCS rules.

Verified GHG emission reductions and removals in the above verification period:

For non-AFOLU projects, use the following table:

Year	Baseline emissions or removals (tCO_2e)	Project emissions or removals (tCO_2e)	Leakage emissions (tCO_2e)	Net GHG emission reductions or removals (tCO_2e)
2012	26,434	4,245	0	22,189
2013	46,874	7,517	0	39,357
2014	27,857	4,486	0	23,371
2015	23,558	3,796	0	19,762

2016	35,379	5,650	0	29,729
2017	55,271	8,867	0	46,403
2018	6457	1109	0	5,348
Total	221,830	35,671	0	186,159

APPENDIX 1: ABBREVIATIONS

BAU	Business As Usual
CA	Corrective Action / Clarification Action
CER	Certified Emission Reduction
CAR	Corrective Action Request
CCIPL	Carbon Check (India) Private Ltd.
CDM	Clean Development Mechanism
CER	Certified Emission Reduction
CL	Clarification Request
CO_2	Carbon Dioxide
CO_{2e}	Carbon Dioxide Equivalent
DOE	Designated Operational Entities
DVR	Draft Validation Report
EB	CDM Executive Board
EF	Emission Factor
FA	Final Approval
FAR	Forward Action Request
FVR	Final validation Report
GHG	Greenhouse gas(es)
GWh	Giga Watt Hour
IPCC	Intergovernmental Panel on Climate Change
MW	Megawatts
MWh	Mega Watt Hour
OSV	On Site Visit

QC/QA	Quality control/ Quality assurance
TA	Technical Area
TR	Technical Review
UNFCCC	United Nations Framework Convention on Climate Change
VCS	Verified Carbon Standard
VVB	Validation / Verification Body

APPENDIX 2: REFERNCE DOCUMENTS

Sl. No.	Title
/01/	a) Monitoring Report version 01 dated 19/11/2021 b) Monitoring Report version 02 dated 24/12/2021 c) Monitoring Report version 2.1 dated 08/01/2022 d) Monitoring report version 2.3 dated 05/04/2022
/02/	Emission reduction calculation sheet version 1.1
/03/	Technical specifications of the following including their nameplates: a) Turbines b) Energy Meters c) Generators
/04/	Calibration certificates of all electricity meters installed (covering the monitoring period), including initial calibration certificates)
/05/	Map of project location showing exact position of HPP with grid connectivity.
/06/	Records of replacement of broken/faulty electricity meter(s), if applicable
/07/	Daily electricity generation log and Monthly electricity generation records of electricity maintained by the power plant operators.
/08/	Monthly electricity sales invoice, issued to the fed to the Vietnam national grid.
/09/	Single line diagram showing the electricity generation, transmission, evacuation and metering system.
/10/	Training records of workers and site employees
/11/	Commissioning certificate of the HEPP.
13	Evidence for start date of project activity
/15/	Power purchase agreement between PP and Vietnam national grid.
/16/	Topographical surveys/maps, satellite pictures used for measurement of Apj.
/17/	Surface Area calculation by third party consultant for the reservoir of the project.
/18/	Name Plate of equipment generating values for CAP_{BJ}
/19/	Monitoring records for the environmental and socio economic impact
/20/	copy of decision 25/2007/QD-BKHCN
/21/	Declaration by PP to avoid double counting for the project activity.
/B01/	VCS Requirements a. VCS Standard (v4.1, dated 19/09/2019) b. VCS Program Guide (v4.0, dated 19/09/2019) c. VCS Validation and Verification Manual version (v3.2, dated 19/10/2016) d. Registration & Issuance Process (v4.0, dated 19/09/2019) e. VCS Programme Definitions version (v4.1, dated 19/09/2019) f. VCS PD template version 4.0
/B02/	Applied baseline and monitoring methodology: ACM0002: Large-scale Consolidated Methodology: Grid-connected electricity generation from renewable sources, Version 11.0

/B03/	Registered CDM PDD version 03.6 dated 19/06/2012 and VCS PD dated 04/12/2012 version 1.1
/B04/	• Previous Monitoring report: (version 3.2 dated 27/06/2012) • Previous verification report: (version 1.2 dated 14/08/2012)
/B05/	1. VCS MR Template (Version 04) 2. Verification report Template (Version 04)

APPENDIX 3: COMPETENCY CERTIFICATE

Carbon CHECK

Carbon Check (India) Private Ltd.

Amit Anand

has been qualified as per CCIPL's internal qualification procedures, in accordance with requirements of Accreditation Standard (version 07.0):

For following functions:

Validator	☒	**Team Leader**	☒	**Technical reviewer**	☒
Verifier	☒	**Technical Expert**	☒	**Local Assessor**[1]	☒

In the following Technical Areas:

TA 1.1	☒	**TA 4.1**	☐	**TA 9.1**	☐	**TA 13.1**	☒
TA 1.2	☒	**TA 5.1**	☐	**TA 9.2**	☐	**TA 13.2**	☐
TA 3.1	☒	**TA 5.2**	☐	**TA 10.1**	☐	**TA 14.1**	☒

Vikash K. S.

Mr. Vikash Kumar Singh
Compliance Officer

Date of Approval
24/12/2021

Valid Till
23/12/2022

Revision History of the Document

01/03/2020[2]	Interim Revision for office address change
01/09/2020	Interim Revision for CCIPL logo change
24/12/2020	Annual Revision
24/12/2021	Annual Revision

[1] India and South Africa

[2] Please refer to previous version of competency certificates for the revision history.

CARBON CHECK (INDIA) PRIVATE LIMITED
CIN: U74930DL2012PTC232495
Regd. Off: 2071/38, 2nd Floor, Naiwala, Karol Bagh, New Delhi - 110005
Corporate off: Unit No. 1701, Logix City Centre Office Tower, Plot No. BW-58, Sector-32 Noida, Uttar Pradesh
Tel: +91 120 4373114 | **URL:** www.carboncheck.co.in | **e-mail:** info@carboncheck.co.in

Carbon CHECK

Carbon Check (India) Private Ltd.

Ms. Aparna Chaudhary

has been qualified as per CCIPL's internal qualification procedures, in accordance with requirements of Accreditation Standard (version 07.0):

For following functions:

Validator	☒	Team Leader	☒	Technical reviewer	☐
Verifier	☒	Technical Expert	☒	Local Assessor[1]	☒

In the following Technical Areas:

TA 1.1	☐	TA 3.1	☒	TA 9.1	☐	TA 13.1	☐
TA 1.2	☒	TA 4.1	☐	TA 9.2	☐	TA 13.2	☐
TA 2.1	☐	TA 5.1	☐	TA 10.1	☐	TA 14.1	☐

Vikash K. Singh

Mr. Vikash Kumar Singh
Compliance Officer

Mr. Amit Anand
CEO

Date of Approval
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01/09/2020	Interim Revision for CCIPL logo change
24/12/2020	Annual Revision
29/11/2021	Revision in response to qualification as Team Leader and Technical Expert

[1] India

[2] Please refer to previous version of competency certificates for the revision history.

CARBON CHECK (INDIA) PRIVATE LIMITED
CIN: U74930DL2012PTC232495
Regd. Off: 2071/38, 2nd Floor, Naiwala, Karol Bagh, New Delhi - 110005
Corporate off: Unit No. 1701, Logix City Centre Office Tower, Plot No. BW-58, Sector-32 Noida, Uttar Pradesh
Tel: +91 120 4373114 | **URL:** www.carboncheck.co.in | **e-mail:** info@carboncheck.co.in

Carbon CHECK

Carbon Check (India) Private Ltd.

Ms. Indumathi. C

has been qualified as per CCIPL's internal qualification procedures, in accordance with requirements of Accreditation Standard (version 07.0):

For following functions:

Validator	☐	**Team Leader**	☒	**Technical reviewer**	☒
Verifier	☐	**Technical Expert**	☒	**Local Assessor**[1]	☒

In the following Technical Areas:

TA 1.1	☒	**TA 4.1**	☐	**TA 9.1**	☐	**TA 13.1**	☒
TA 1.2	☒	**TA 5.1**	☐	**TA 9.2**	☐	**TA 13.2**	☒
TA 3.1	☒	**TA 5.2**	☐	**TA 10.1**	☐	**TA 14.1**	☐

Vikash K. Singh

Mr. Vikash Kumar Singh
Compliance Officer

Mr. Amit Anand
CEO

Date of Approval
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Revision History of the Document

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01/09/2020	Interim Revision for CCIPL logo change
24/12/2020	Annual Revision
24/12/2021	Annual Revision

[1] India.

[2] Please refer to previous version of competency certificates for the revision history.

CARBON CHECK (INDIA) PRIVATE LIMITED
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Tel: +91 120 4373114 | **URL:** www.carboncheck.co.in | **e-mail:** info@carboncheck.co.in

APPENDIX 4: CLARIFICATION REQUESTS, CORRECTIVE ACTION REQUESTS AND FORWARD ACTION REQUEST

Finding	CAR 01		
Classification	☒ CAR	☐ CL	☐ FAR
Description of finding (DOE)	*As per the VCS MR template version 1 under section 1.1 "A summary description of the implementation status of the technologies/ measures (e.g., plant, equipment, process, or management or conservation measure) included in the project."* *However, it has been observed by the verification team that the information about all the equipment is not mentioned in the MR section 1.1. PP is requested to provide complete information of all the equipment used.*		
Corrective Action or clarification #1 *(PP shall write a detailed and clear corrective action or further information for clarification as per finding)*	*A summary of equipment used has been provided in section 1.1 as well as a reference to section 3.1 which has more detailed information.*		
DOE Assessment #1 *The assessment shall encompass all open issues in the finding. In case of non-closure, additional corrective action and DOE assessments (#2, #3, etc.) shall be added.*	*PP has provided the information of all the equipment used under the section 1.1 of the VCS MR and also the details of the equipment used is provided under section 3.1 of the revised MR.* *Hence this finding is closed.*		
Conclusion *Tick the appropriate checkbox*	☐ *To be checked during the next periodic verification* ☐ *Outstanding finding (not closed)* ☒ *The finding is closed*		

Finding	CAR 02		
Classification	☒ CAR	☐ CL	☐ FAR
Description of finding (DOE)	*Under section 2.1 of MR, PP has mentioned the details of the Environmental Impact assessment in compliance with the registered PD. PP is requested to submit the monitoring records for this monitoring period showing that all the mitigation measures had been taken into account during this monitoring period.*		

Finding	*CAR 02*
Corrective Action or clarification #1 *(PP shall write a detailed and clear corrective action or further information for clarification as per finding)*	*A third party monitors and prepares reports of environmental compliance on a regular basis. Any identified issues are recorded and tracked until they are resolved, and any mitigation measures are recorded. Reports and selected test results have been provided to the DOE.*
DOE Assessment #1 *The assessment shall encompass all open issues in the finding. In case of non-closure, additional corrective action and DOE assessments (#2, #3, etc.) shall be added.*	*On the basis of review of the environmental monitoring reports for all the four quarters, environmental protection report 2020_Dak Srong and Test results Dak Srong 2 (Quarter I, II, III, IV) verification team concludes that all the potential environmental and socio-economic impacts of the project were not significant but was temporary due to the project development phase rather than the operational phase of the project. However, project owners have followed the environmental monitoring plan in complies with the registered PD and with all related Vietnamese laws.* *Hence the finding is closed.*
Conclusion *Tick the appropriate checkbox*	☐ *To be checked during the next periodic verification* ☐ *Outstanding finding (not closed)* ☒ *The finding is closed*

Finding	*CAR 03*		
Classification	☒ *CAR*	☐ *CL*	☐ *FAR*
Description of finding *(DOE)*	*Under section 2.2 of the MR the details of LSC have been provided. PP is requested to mention if the project receive any feedback from local stakeholder during this monitoring period. Moreover, PP is requested to submit grievance registered maintained for this monitoring period.*		
Corrective Action or clarification #1 *(PP shall write a detailed and clear corrective action or further information for clarification as per finding)*	*Any grievances from local stakeholders are to be reported through the local authority and measures and status will be included in the environmental monitoring report(s). No grievances have been received by the project. A report from a recent consultation (with community leaders and specifically addressing water usage) is provided that shows no reported issues nor any pending/historical issues.*		
DOE Assessment #1 *The assessment shall encompass all open issues in the finding. In case of non-closure, additional corrective action and DOE assessments (#2, #3, etc.) shall be added.*	*Verification team has reviewed the supporting document namely Water usage consultation 2&2A which is a report for a recent consultation with community leaders and specifically addressing water usage. The report shows that there is no reported issues nor any pending/historical issues and thus confirms that there is no grievance registered for this monitoring period.* *Hence the finding is closed.*		

Finding	CAR 03
Conclusion *Tick the appropriate checkbox*	☐ *To be checked during the next periodic verification* ☐ *Outstanding finding (not closed)* ☒ *The finding is closed*

Finding	CAR 04		
Classification	☒ *CAR*	☐ *CL*	☐ *FAR*
Description of finding *(DOE)*	*As per the filling guideline of the VCS MR template under section 3.1 a summary description of the implementation status of the technologies/ measures (e.g., plant, equipment, process, or management or conservation measure) included in the project." Has to be provided.* *PP is requested to provide the complete operational status including number of equipment used, the capacity of the equipment including the details about the transformer and substation used for the implementation of the project.*		
Corrective Action or clarification #1 *(PP shall write a detailed and clear corrective action or further information for clarification as per finding)*	*Details have been added to section 3.1.*		
DOE Assessment #1 *The assessment shall encompass all open issues in the finding. In case of non-closure, additional corrective action and DOE assessments (#2, #3, etc.) shall be added.*	*On the basis of review of the section 3.1 of the revised MR verification team confirms that the details required in the section as per the VCS MR template has been provided by the PP.* *Hence this finding is closed.*		
Conclusion *Tick the appropriate checkbox*	☐ *To be checked during the next periodic verification* ☐ *Outstanding finding (not closed)* ☒ *The finding is closed*		

Finding	CAR 05		
Classification	☒ *CAR*	☐ *CL*	☐ *FAR*

Finding	CAR 05
Description of finding (DOE)	*Under section 4.2, table for the parameter EGfacility, PP is requested to provide the complete details of the monitoring equipment including* *Make/model: (For all meters)* *Type:* *Accuracy:* *Serial number of main meter 1:* *Serial number of back up meter 1:* *Date of initial calibration:* *Date of previous calibration:* *Validity of calibration* *Moreover, PP is requested to provide the evidence for the calibration frequency of every 3 years as mentioned in the MR.* *However, it has been noted by verification team that the calibration frequency has not been provided in the registered CDM PDD.* *It has been also observed that the measuring equipment mentioned in the registered CDM PDD is bidirectional meters and in MR it is not mentioned. PP is requested to clarify the inconsistency.*
Corrective Action or clarification #1 *(PP shall write a detailed and clear corrective action or further information for clarification as per finding)*	*As there is a lot of information due to the number of meters involved and the duration of the monitoring period, these details have been added in Appendix 1 and a note to refer to this has been added to section 4.2. Also a copy of decision 25/2007/QD-BKHCN has been provided which establishes a 2 year calibration frequency for electricity meters.*
DOE Assessment #1 *The assessment shall encompass all open issues in the finding. In case of non-closure, additional corrective action and DOE assessments (#2, #3, etc.) shall be added.*	*On the basis of review of the revised MR submitted by the PP verification team confirms that PP has provided the reference of appendix 1 under the section 4.2 or the details of the calibration records of all the electricity meters used in the projects.* *Also PP has provided the evidence for the calibration frequency of meters as 2 years.* *Hence the finding is closed.*
Conclusion *Tick the appropriate checkbox*	☐ *To be checked during the next periodic verification* ☐ *Outstanding finding (not closed)* ☒ *The finding is closed*

Finding	CL 01		
Classification	☐ *CAR*	☒ *CL*	☐ *FAR*
Description of finding (DOE)	*Under section 1.5 PP has mentioned first commercial operation date of first unit. PP is requested to clarify what does first unit referred for.*		

Finding	CL 01
Corrective Action or clarification #1 *(PP shall write a detailed and clear corrective action or further information for clarification as per finding)*	*The project consists of three units or turbine/generator sets, each with a capacity of 8 MW and these were put into commercial operation sequentially. However, this has been changed as the additional information has no relevance – the date when the project first started exporting and selling electricity is the commercial operations date for the whole project.*
DOE Assessment #1 *The assessment shall encompass all open issues in the finding. In case of non-closure, additional corrective action and DOE assessments (#2, #3, etc.) shall be added.*	*PP has removed the reference first in the revised MR as provided in the section 1.5 of the previous version of MR to avoid any confusion. Moreover, PP has explained about the commercial operation date for the project as mentioned above. The same is deemed acceptable by the verification team and thus the finding is closed.*
Conclusion *Tick the appropriate checkbox*	☐ *To be checked during the next periodic verification* ☐ *Outstanding finding (not closed)* ☒ *The finding is closed*

Finding	CL 02		
Classification	☐ *CAR*	☒ *CL*	☐ *FAR*
Description of finding *(DOE)*	*Under section 4.2, table for the parameter CAP BJ, monitoring frequency is measured as any changes record, however in the registered CDM PDD it is mentioned as yearly. PP is requested to clarify the inconsistency.* *.*		
Corrective Action or clarification #1 *(PP shall write a detailed and clear corrective action or further information for clarification as per finding)*	*This has been corrected in the MR.*		
DOE Assessment #1 *The assessment shall encompass all open issues in the finding. In case of non-closure, additional corrective action and DOE assessments (#2, #3, etc.) shall be added.*	*PP has made correction in the section 4.2 for the parameter CAP BJ, and the monitoring frequency is mentioned as yearly which is now consistent with the registered PD. Hence the finding is closed.*		
Conclusion *Tick the appropriate checkbox*	☐ *To be checked during the next periodic verification* ☐ *Outstanding finding (not closed)* ☒ *The finding is closed*		

Finding	*CL 03*		
Classification	☐ *CAR*	☒ *CL*	☐ *FAR*
***Description of finding** (DOE)*	*Verification team has observed that the actual generation in the year 2017 was higher than the estimated. Please clarify. Moreover, generation in the following months are higher. Oct-Dec 2013, Oct-Dec 2016, Jan 2017 and Nov-Dec 2017*		

Corrective Action or clarification #1 *(PP shall write a detailed and clear corrective action or further information for clarification as per finding)*	*The estimated output of hydropower electric power plants, particularly for run of the river facilities, are difficult to model with high accuracy. However, over a longer period of time, the performance tends to approach a good estimate.* *Two major factors impact this project –* *1) Variations in precipitation* *2) A large hydropower project upstream* *According to the Hydropower Modeling Challenges (NREL, 2017: Hydropower Modeling Challenges (nrel.gov))):* *"Run-of-river hydropower generators exist alongside rivers and do not contain a large reservoir to store or regulate the flow of the adjacent river. Sometimes these facilities have a small storing capacity that can be used to modify the generation profile to a small degree according to system load or electricity prices; however, typically they generate electricity according to the water flow. The provision of electricity generation as a second-order interest—following environmental, operational, and regulatory constraints—restricts the degree of flexibility that these facilities can provide* *And again:* *"Hydropower operations are at the mercy of ever-changing hydrological conditions. Reservoirs are impacted by water inflows—which may include natural inflows as well as the discharge of upstream hydropower generation or pumping units"* *Taking the end of 2016, early 2017 as an example, the Dak Srong hydropower project had increased generation due to the increased precipitation as well as regulated discharge from the up river project - An Khe Kanak. From this article (Central provinces strengthen efforts to cope with widespread flooding - Nhan Dan Online):* *"According to Vietnam Electricity (EVN), 13 hydropower reservoirs have commenced with regulated water discharge, including Quang Tri, A Vuong, Tranh River 2, Vinh Son A, Bung River 4, Ba Ha River, Hinh River, Ka Nak , An Khe, Buon Kuop, Srepok 3, Don Duong and Dai Ninh. EVN also sent dispatches to relevant units to deploy prevention plans to deal with flooding"* *2013 and 2017 also saw floods in Gia Lai:* *Vietnam – Floods (ECHO Daily Flash of 19 November 2013) - Viet Nam | ReliefWeb*

Finding	*CL 03*
	Situation Update No. 2: Typhoon Damrey, Viet Nam - Friday, 10 November 2017, 18:00 hrs (UTC+7) - Viet Nam | ReliefWeb *In the case of this project, while 2017 saw a plant load factor of 46% - which while higher than the 44% as per the registered PDD. However, this is balanced by significantly lower performance in other years (as low as 20% in 2015) – leading to an implied PLF of 30% over the six year monitoring period.*
DOE Assessment #1 *The assessment shall encompass all open issues in the finding. In case of non-closure, additional corrective action and DOE assessments (#2, #3, etc.) shall be added.*	*On the basis of PP response above it has been observed that the major factor which impact the project are variation in precipitation and a large hydropower upstream.* Moreover, by reviewing the *National Renewable Energy Laboratory (NREL) 2017 "Run-of-river hydropower generators exist alongside rivers and do not contain a large reservoir to store or regulate the flow of the adjacent river. Sometimes these facilities have a small storing capacity that can be used to modify the generation profile to a small degree according to system load or electricity prices; however, typically they generate electricity according to the water flow. The provision of electricity generation as a second-order interest—following environmental, operational, and regulatory constraints—restricts the degree of flexibility that these facilities can provide" However the hydropower depends on the hydrological and climatic conditions and thus impacted by the natural inflows and the upstream discharges that effects the generation.* *The early 2017 has an increased generation due to higher precipitation and thus higher flow which also occurs due to the regulated discharge from the upriver projects.* *In addition to that there was flood situations in the year 2013 to 2017 in Gia Lai as checked from the :* Vietnam – Floods (ECHO Daily Flash of 19 November 2013) - Viet Nam | ReliefWeb Situation Update No. 2: Typhoon Damrey, Viet Nam - Friday, 10 November 2017, 18:00 hrs (UTC+7) - Viet Nam | ReliefWeb Thus, the PLF for the project implies to 30% in the monitoring period as in 2017 it was higher than the 44% as in registered PD while in other years it was as low as as 20% in 2015. Based on the above assessment the change in the actual generation as compared to estimated generation for the above mentioned years are deemed acceptable to the verification team. Hence the finding is closed.

Finding	*CL 03*
Conclusion *Tick the appropriate checkbox*	☐ *To be checked during the next periodic verification* ☐ *Outstanding finding (not closed)* ☒ *The finding is closed*

APPENDIX 5: ASSESSMENT OF MONITORING PARAMETERS FIXED EX-ANTE

Parameter	Description	Value	Unit	Source	Assessment
EFgrid,CM,y	CO2 emission factor in year y Combined margin CO2 emission factor for grid connected power generation in year y calculated using "Tool to calculate the emission factor for an electricity system" (version 01).	0.5764	tCO_2/MWh	As per GEF report from Vietnam DNA "Study, Definition of Viet Nam Grid Emission Factor, 2009	The parameter is used for the calculation of the baseline emission the project activity. The value is consistent with the registered CDM PD and VCS PD/B03/ and fixed ex-ante for the project activity.
EF_{Res}	Default emission factor for emission from reservoirs	90	Kg CO_2/MWh	Decision by EB 23	The value is consistent with the registered CDM and VCS PD/B03/ and fixed ex-ante for the project activity as per the applied methodology. The value is used to calculate the baseline emissions.